Out-of-Distribution Detection Inverts Under Domain-Adversarial Training

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Editors: Under Review for MIDL 2027

Abstract

Background. Domain-adversarial invariance training is widely recommended for removing acquisition shortcuts from medical imaging models. Its effect on the out-of-distribution (OOD) monitoring that such models rely on at deployment has not been measured.

Methods. We swept the adversarial weight λ_{adv} across seven values with three to five seeds each (27 runs) on ISIC 2019 (dermoscopy, 25,331 images) and PAD-UFES-20 (clinical photographs, 2,298 images), holding everything else fixed. At every λ_{adv} we measured domain leakage by linear probe, in-distribution balanced accuracy, calibration on in-distribution and out-of-domain data, OOD detection with five scoring rules, and zero-shot cross-domain diagnostic accuracy on a patient-level held-out PAD partition. We repeated the sweep with plain single-encoder adversarial training and with four non-adversarial invariance objectives, and validated on Fitzpatrick17k, a third dataset never seen by the adversary.

Results. At the smallest non-zero weight tested, OOD detection fell from AUROC 0.843 to 0.513; by $\lambda_{\text{adv}} = 2$ it had **inverted below chance to** 0.427 (95% CI $[0.414, 0.439]$; all five seeds individually below 0.5), ranking images from a clinic the model had never seen as more in-distribution than its own test data, throughout the score distribution rather than only in the tails. All five scoring rules collapsed together, the inversion held on the third dataset (0.391 at $\lambda_{\text{adv}} = 2$ against 0.764 at $\lambda_{\text{adv}} = 0$, the same model and data) and survived shrinkage estimation. The collapse was a cliff rather than a trade-off: no operating point combined invariance with detection. It was invisible to routine monitoring: domain leakage fell to near its chance floor (0.915 → 0.571, floor 0.5), in-distribution calibration was unchanged, and in-distribution balanced accuracy did not degrade (0.692 → 0.707). Because 60.0% of the ISIC test split shares a lesion with training, we repeated the endpoints under a lesion-disjoint split at five seeds. Everything holds: leakage 0.927 → 0.555, in-distribution ECE unchanged at 0.181, out-of-domain ECE 0.219 → 0.717, and the inversion 0.808 → 0.403 (95% CI $[0.380, 0.425]$, all five seeds individually below 0.5) on `pad_heldout` and 0.717 → 0.402 on Fitzpatrick17k. In-distribution balanced accuracy again does not degrade: the paired per-seed difference is -0.023 with a 95% CI of $[-0.063, +0.017]$. Removing the lesion overlap costs 0.20 of balanced accuracy on its own (0.692 → 0.496), an order of magnitude more than any effect of λ_{adv}. Confidence on never-seen-domain images rose from 0.489 to 0.930, overtaking in-distribution confidence, while cross-domain accuracy did not improve at any λ_{adv} and predictions collapsed toward the benign majority class (actinic keratosis recall 0.245 → 0.016, $n = 202$). The failure reproduced in plain single-encoder gradient-reversal training but not in moment-matching objectives, which reduced leakage comparably without collapsing any detector. In a same-modality shift between two hospitals the collapse reproduced (k-NN 0.658 → 0.486) but the inversion did not: with matched baseline detectability, Mahalanobis fell only to 0.683. After six candidate explanations were tested and refuted, the cause was identified and satisfied three pre-registered criteria: invariance is obtained by displacing out-of-domain images into the

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majority class's region of the latent space, so the fraction of them whose nearest centroid is nevus rises 0.332 → 0.601 while the in-distribution fraction is flat, the inversion is confined to that subset (0.807 → 0.282 against 0.861 → 0.644), and their distance to the nevus centroid falls 49.8 → 17.5 against an unchanged class-matched reference.

Conclusions. Across imaging modality, domain-adversarial training does not merely weaken out-of-distribution monitoring; it reverses it, while every routinely monitored metric improves. The cause is that invariance is obtained by the one move that defeats a density-based novelty score. Between two hospitals sharing a modality the collapse persists and the reversal does not, so the two should be claimed separately. OOD detection and out-of-domain calibration should be reported as a function of achieved invariance rather than at a single operating point.

Keywords: domain-adversarial training; out-of-distribution detection; calibration; distribution shift; skin lesion classification; model monitoring

1. Introduction

Deep models for skin lesion classification are known to exploit acquisition artefacts rather than pathology: ruler marks, ink annotations, imaging hardware and site identity all correlate with diagnosis in public benchmarks (Winkler et al., 2019; Bissoto et al., 2020; Daneshjou et al., 2022). The standard remedy is to make the representation invariant to the acquisition domain, most often through domain-adversarial training with a gradient reversal layer (Ganin et al., 2016; Bousmalis et al., 2016), and its stated purpose is safe deployment across sites. Deployment across sites also relies on a second mechanism: out-of-distribution monitoring, by which a system flags inputs unlike its training data so a human reviews them rather than the model acting alone. This paper asks what invariance training does to that mechanism.

It reverses it. Sweeping the adversarial weight in a controlled setting, we find that out-of-distribution detection does not merely weaken as domain information is removed — it passes through chance and inverts. At the weights that achieve invariance, a model asked which inputs are unlike its training data answers with its own test split and ranks images from a clinic it has never seen as the more routine of the two. AUROC falls from 0.843 to 0.427, with a 95% confidence interval of $[0.414, 0.439]$ lying entirely below chance, across every scoring rule we tried.

Degradation would be unsurprising: a representation stripped of domain information should be worse at recognising a domain, and chance is what that argument predicts. Inversion is not, and accounting for it took six refuted explanations and a failed pre-registered synthetic reproduction before a mechanism held — invariance is obtained by displacing out-of-domain images into the region the majority class occupies, which a density-based novelty score reads as familiarity.

What makes the reversal consequential rather than merely odd is that nothing in a standard evaluation reveals it. Over the same range the model becomes more confident on unseen-domain images than on its own held-out data and its predictions collapse toward the benign majority class, while in-distribution accuracy does not fall, in-distribution calibration does not change, and the domain leakage probe — the very metric the technique is adopted to improve — improves exactly as intended. The technique disables the cross-site safety machinery while every monitored number moves in the direction its user is hoping for.

Contributions.

1. A controlled intervention: a seven-point sweep of the adversarial weight, 27 training runs with everything else held fixed, isolating invariance pressure from architectural and data confounds.

2. The inversion: OOD detection passes through chance and reverses, as a cliff rather than a trade-off curve, across five scoring rules, on a third dataset the adversary never saw, and robustly to shrinkage estimation and tail truncation.

3. The invisibility: in-distribution accuracy, in-distribution calibration and the domain leakage probe all fail to reveal it, under both an image-level and a lesion-disjoint ISIC split at five seeds, and the leakage probe does not mediate it, so it cannot serve as a proxy either.

4. Scope: the failure reproduces in plain single-encoder adversarial training, so it is not an artefact of the factorized architecture; and it does not appear under moment-matching objectives that reduce leakage along the same axis, so it belongs to adversarial optimisation rather than to invariance.

5. A mechanism with predictive content, satisfying three pre-registered criteria and replicating on a second representation, after six alternative accounts and a pre-registered synthetic reproduction were tested and refuted.

6. A reporting protocol that makes the failure visible at near-zero cost.

2. Related work

Shortcut learning in dermatology. Ink markings shift melanoma predictions (Winkler et al., 2019), classifiers keep competitive ISIC accuracy with the lesion masked out (Bissoto et al., 2020), and performance degrades on darker skin tones (Daneshjou et al., 2022). These motivate invariance-based remedies.

Invariant representation learning. Domain-adversarial training (Ganin et al., 2016) and domain separation networks (Bousmalis et al., 2016) remove domain information by adversarial or orthogonality constraints. **The architecture used here is equivalent to a domain separation network and is not a contribution of this paper**; it is a vehicle for the intervention.

Covariate shift versus semantic novelty. Full-spectrum OOD detection (Yang et al., 2023) argues the literature conflates two shifts, and proposes a taxonomy — training ID, *covariate-shifted ID*, near-OOD and far-OOD — in which covariate-shifted in-distribution samples should be tolerated rather than flagged, since rejecting them contradicts generalization; OpenOOD v1.5 (Zhang et al., 2023) adopts the same setting at ImageNet scale. Our contrast falls in the covariate-shifted-ID cell, so a reader of (Yang et al., 2023) might regard a model that accepts these images as behaving correctly. Section 5 argues tolerance is safe only with generalization, and here it is not.

Calibration under adaptation, and the reverse direction. Calibration degrades under shift for ordinary training (Ovadia et al., 2019), and domain adaptation methods specifically *trade* calibration for accuracy on the target domain, for which a post-hoc correction exists (Wang et al., 2020). **The calibration degradation in Section 4.4 is therefore not a new phenomenon.** New here are its dose-dependence on the adversarial weight, its invisibility to in-distribution calibration, and — separately from calibration — the collapse and inversion of OOD detection, which a confidence correction does not address. In the forward direction, enforcing multi-domain calibration yields invariance and provable freedom from certain spurious correlations (Wald et al., 2021); we find the reverse implication does not hold (Section 4.4).

3. Materials and methods

3.1. Data

ISIC 2019 (Tschandl et al., 2018): 25,331 dermoscopic images, eight diagnostic classes (MEL, NV, BCC, AK, BKL, DF, VASC, SCC), CC BY-NC 4.0. **PAD-UFES-20** (Pacheco et al., 2020): 2,298 smartphone clinical photographs, six classes mapping into the same label space (DF and VASC absent), CC BY 4.0. **Fitzpatrick17k** (Groh et al., 2021): 3,887 of 16,577 images obtained; annotations CC BY-NC-SA 3.0, images web-scraped under mixed copyright, used for evaluation only and not redistributed.

Images were cropped to a soft lesion-centred bounding box via Otsu thresholding on the deviation from the border median, expanded by a 0.30 margin with a 0.70 minimum crop ratio, and resized to 224×224.

3.2. Splits and the PAD held-out set

ISIC was split 60/20/20 into train/validation/test, stratified by label (`random_state` = 42).

PAD appears in training and would otherwise also be the OOD set. The classification loss never sees PAD labels (`ignore_index`= -1), but PAD images enter the context and adversarial branches. To remove the resulting circularity we partitioned PAD *at patient level* into `pad_adv` (1,582 images, 961 patients), visible to those branches during training, and `pad_heldout` (716 images, 412 patients), never seen in any form. No patient and no lesion appears in both. **All OOD and cross-domain numbers reported in the main text use `pad_heldout`.** An earlier draft of this work reported 0.413 at $\lambda_{\text{adv}} = 2$; that value was scored on `pad_full`, which includes the 1,582 images the adversarial branch saw during training. Fitzpatrick17k provides a second never-adversarial domain.

3.3. Architecture and objective

A dual-encoder model with EfficientNet-B3 backbones produces a lesion latent $z_\ell \in \mathbb{R}^{16}$ (batch-normalised, denoted z_ℓ^{norm}) and a context latent $z_c \in \mathbb{R}^{64}$. Diagnosis uses z_ℓ^{norm} only, through a linear head. The objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{ctx}}\mathcal{L}_{\text{ctx}} + \lambda_{\text{adv}}\,\mathcal{L}_{\text{adv}} + \lambda_{\text{orth}}\mathcal{L}_{\text{orth}}, \tag{1}$$

with $\mathcal{L}_{\text{cls}}$ on ISIC only, $\mathcal{L}_{\text{ctx}}$ a domain cross-entropy on z_c, $\mathcal{L}_{\text{adv}}$ a domain cross-entropy on z_ℓ^{norm} behind a gradient reversal layer, and $\mathcal{L}_{\text{orth}}$ a squared-cosine penalty between the two latents ($\lambda_{\text{orth}} = 1$ throughout).

AdamW, base learning rate 10^{-4}, adversary learning rate multiplier 30, classifier multiplier 0.2, 40 epochs, batches balanced across domains, checkpoint selected on validation accuracy.

3.4. The intervention

$\lambda_{\text{adv}} \in \{0, 0.25, 0.5, 1, 2, 4, 8\}$; five seeds at $\lambda_{\text{adv}} \in \{0, 2, 8\}$ and three at the interior points, for 27 runs. Everything else is held fixed, so λ_{adv} is the only manipulated variable.

3.5. Measurements

Domain leakage. Logistic probe on frozen features predicting the source dataset, reported as **balanced accuracy, whose chance floor is** 0.5 **for this two-domain problem**. Earlier work on this dataset pair reports plain accuracy against a majority floor of 0.688; the two scales must not be compared to each other. Both are stated where relevant.

OOD detection. Maximum softmax probability (Hendrycks and Gimpel, 2017), free energy (Liu et al., 2020), maximum cosine to a class centroid, class-conditional Mahalanobis with shared covariance (Lee et al., 2018), and k-NN distance (Sun et al., 2022) ($k \in \{10, 50, 200\}$), all fit on the ISIC training split alone. **These five rules are not independent evidence**: the classifier is a linear map from z_ℓ^{norm}, so all five are functions of the same 16-dimensional vector. Independent evidence comes from the pre-projection backbone (Section 4.5).

Calibration. Expected calibration error, 15 bins, on ISIC test and on `pad_heldout` restricted to the six shared classes. **Out-of-domain ECE here is arithmetically close to mean confidence minus accuracy**, so the confidence, accuracy and ECE results in Section 4.4 are three views of one phenomenon rather than three findings.

Uncertainty. Intervals on AUROC come from a stratified bootstrap, 2,000 resamples, percentile method, fixed in a pre-registration before the values were computed. Each draw resamples **patients** from `pad_heldout` and, independently, images from the ISIC test split, because AUROC depends on both score distributions and resampling one side understates the interval; within a draw the same resampled sets serve every seed. We report the interval on the five-seed mean and the number of seeds whose own interval clears 0.5. Fitzpatrick17k carries no patient identifiers, so its bootstrap is over images and its interval is likely anti-conservative.

Cross-domain diagnosis. Zero-shot on `pad_heldout` over the six shared classes; PAD labels were never used for $\mathcal{L}_{\text{cls}}$.

Table 1: The intervention. Mean ± s.d. across seeds. Leakage is balanced accuracy (chance 0.5); cross-domain accuracy is balanced over the six classes shared with PAD-UFES (chance 0.167). Arrows mark the direction a practitioner would call favourable. The $\lambda_{\text{adv}} = 0.25$ row previously reported three-seed means for leakage, in-distribution accuracy and Fitzpatrick AUROC while declaring $n = 5$; those three are corrected here. † marks cells still at three seeds: the $\lambda_{\text{adv}} = 0.25$ OOD AUROC, and its standard deviations, are being recomputed, and the $\lambda_{\text{adv}} = 1$ row likewise uses three of the five available seeds.

λ_{adv}	n	Leakage ↓	ID bal. acc ↑	ID ECE	OOD ECE	OOD AUROC	Fitz. AUROC	X-dom bal. acc
0	5	0.915 ± 0.021	0.692	0.107	0.247	$\mathbf{0.843 \pm 0.023}$	0.764 ± 0.038	0.291 ± 0.016
0.25	5†	0.571	0.692	0.101	0.684	$\mathbf{0.513 \pm 0.018}$†	0.460	0.291 ± 0.020
0.5	3	0.555 ± 0.029	0.701	0.096	0.724	0.465 ± 0.015	0.449 ± 0.025	0.270 ± 0.034
1	3†	0.569 ± 0.010	0.701	0.097	0.733	0.455 ± 0.036	0.457 ± 0.088	0.261 ± 0.016
2	5	0.584 ± 0.027	0.707	0.099	0.746	$\mathbf{0.427 \pm 0.025}$	0.391 ± 0.068	0.249 ± 0.030
4	3	0.548 ± 0.035	0.686	0.100	0.726	0.455 ± 0.039	0.431 ± 0.036	0.276 ± 0.025
8	5	0.625 ± 0.045	0.679	0.099	0.713	0.494 ± 0.027	0.474 ± 0.060	0.269 ± 0.023
EffNet-B3 control		0.802	0.683	0.096	—	0.726 ± 0.021	0.649	0.298 ± 0.010
ResNet-50 baseline		0.979	0.655	0.101	—	0.868 ± 0.019	0.994	0.391

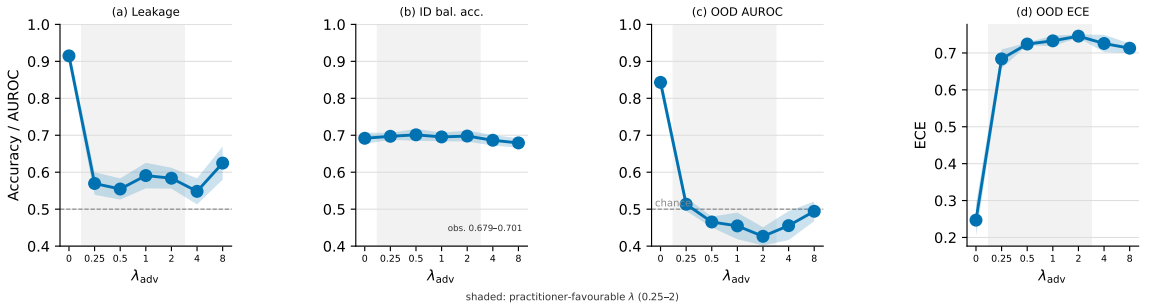


Figure 1: **The intervention, four views, one axis.** Leakage falls to its chance floor and in-distribution balanced accuracy does not degrade — both favourable. Over the same range OOD AUROC collapses through chance and inverts, and out-of-domain calibration error triples. A developer monitoring the first two panels would see nothing wrong.

4. Results

4.1. The intervention

Table 1 reports every measurement as a function of λ_{adv}, the only manipulated variable. Figure 1 shows the four quantities that matter most on a shared axis.

Read across a single row and nothing is wrong. Read down the columns and two of them improve while two collapse.

4.2. Invariance is achieved, and in-distribution metrics do not warn

Domain leakage falls from 0.915 to 0.571 between $\lambda_{\text{adv}} = 0$ and $\lambda_{\text{adv}} = 0.25$, against a chance floor of 0.5. By the smallest non-zero weight tested, the lesion representation is very nearly uninformative about the acquisition source: the intended effect of the method, obtained essentially in full.

It is obtained without an in-distribution cost. Balanced accuracy on the ISIC test split moves from 0.692 to 0.707 over the useful range — we claim no improvement, the difference being small relative to seed variance — and in-distribution expected calibration error is flat at ≈ 0.10 at every λ_{adv}. A practitioner tracking accuracy, in-distribution calibration and the leakage probe would conclude that the method had worked exactly as advertised.

4.3. Out-of-distribution detection collapses, and inverts

Over the same interval, Mahalanobis AUROC against the held-out PAD partition falls from 0.843 to 0.513 and reaches 0.427 at $\lambda_{\text{adv}} = 2$. Two features matter more than the size of that drop.

The fall to chance is a cliff; the inversion develops after it. Almost the whole descent happens in one step, 0.843 to 0.513 between $\lambda_{\text{adv}} = 0$ and $\lambda_{\text{adv}} = 0.25$ (three seeds; the five-seed value is being recomputed). The sub-chance region is reached only at $\lambda_{\text{adv}} \geq 0.5$, after which the value moves between 0.427 and 0.494 with no ordering a practitioner could tune against. So detection collapses to chance at the smallest non-zero weight tested and inverts under further pressure; there is no operating point at which invariance has been obtained and detection survives.

It goes below chance. At $\lambda_{\text{adv}} = 2$ the detector ranks images from a clinic the model has never seen as *more* in-distribution than its own test split. A patient-clustered bootstrap puts the 95% interval at $[0.414, 0.439]$ with all five seeds' individual intervals below 0.5; the same holds for k-NN (0.440, $[0.429, 0.452]$, 5/5) and, one seed short, cosine (0.453, $[0.441, 0.464]$, 4/5). It is not a tail artefact: removing the pooled 90th, 95th or 99th percentile leaves AUROC at 0.412, 0.416 and 0.424 against 0.427 for the full set, and the medians themselves invert (ISIC 13.8 versus `pad_heldout` 10.0).

All five rules move together — MSP 0.970 $\rightarrow$ 0.437, energy 0.983 $\rightarrow$ 0.441, cosine 0.901 $\rightarrow$ 0.435, k-NN 0.941 $\rightarrow$ 0.424, stable across $k \in \{10, 50, 200\}$ — but as noted in the methods they are five views of one 16-dimensional representation; the independent replications are the backbone (Section 4.5) and the single-encoder model (Section 4.7).

Three artefact explanations fail. *Covariance estimation:* Ledoit–Wolf shrinkage leaves $\lambda_{\text{adv}} = 2$ at 0.427, and across float32 and float64, with and without shrinkage, at one, four and eight BLAS threads, the largest AUROC deviation is 9.4×10^{-4}; effective dimensionality barely moves (participation ratio 4.879 $\rightarrow$ 4.500 while leakage falls 0.914 $\rightarrow$ 0.553 in the same three runs). *An over-strong adversary:* the adversary runs at 30$\times$ the base learning rate, and retraining at 10$\times$ and 100$\times$ keeps the inversion — 0.418 ± 0.011 at 10$\times$ with all three seeds below 0.5, against 0.432 ± 0.024 at 30$\times$ and 0.482 ± 0.020 at 100$\times$ (one seed at 0.503) — so weakening it tenfold deepens the inversion rather than removing it. *The adversary's own domain:* on Fitzpatrick17k, absent from every training branch, AUROC falls 0.764 $\rightarrow$ 0.391 (95% CI $[0.386, 0.396]$, five of five seeds below 0.5; bootstrapped over

images, so likely anti-conservative). For context a ResNet-50 baseline reaches 0.994 on those images and the EfficientNet-B3 single-encoder control 0.649; the controlled comparison is the $\lambda_{\text{adv}} = 0$ row, which differs from $\lambda_{\text{adv}} = 2$ only in the adversarial weight.

4.4. The model becomes confidently wrong on unseen domains

At $\lambda_{\text{adv}} = 0$ the model is appropriately hesitant on unfamiliar data: mean maximum-softmax confidence is 0.915 on the ISIC test split and 0.489 on `pad_heldout`. By $\lambda_{\text{adv}} = 0.25$ that gap has closed (0.913 versus 0.898), and by $\lambda_{\text{adv}} = 2$ it has inverted — the model is *more* confident on images from a clinic it has never seen (0.930) than on its own held-out test set (0.910).

Confidence rises while accuracy falls. Cross-domain diagnostic accuracy does not improve at any λ_{adv}: balanced accuracy spans 0.249–0.291 against a six-class floor of 0.167, no setting exceeds the 0.291 obtained at $\lambda_{\text{adv}} = 0$, and every setting is below the 0.298 of the non-adversarial EfficientNet-B3 control. Plain accuracy falls 0.245 → 0.184, beneath the 0.385 majority-class rate throughout.

Macro AUC is roughly preserved (0.605 → 0.581), so discrimination survives while the decision rule does not. Predictions collapse toward the benign majority class: nevus predictions rise from 113 to 374 of 716 images, and **actinic keratosis recall falls from** 0.245 **to** 0.016 (49/202 → 3/202 images, mean over five seeds). Squamous cell carcinoma moves the same way (0.088 → 0.021, 6/66 → 1/66) but the count is too small to carry a claim on its own, and melanoma ($n = 9$) supports none.

The resulting out-of-domain calibration error triples, from 0.247 to 0.746, while in-distribution ECE stays flat at ≈ 0.10. That adapted models are miscalibrated on the target domain is known (Wang et al., 2020); what the sweep adds is that the degradation is dose-dependent on the adversarial weight and leaves in-distribution calibration untouched, so no in-distribution measurement reveals it. We stress that confidence, accuracy and calibration here are three views of a single phenomenon rather than three independent results: in this setting out-of-domain ECE is arithmetically close to mean confidence minus accuracy.

4.5. Leakage does not mediate the collapse

The natural hypothesis is that removing domain information is what breaks detection. Four results rule it out, the last of them outside dermatology.

First, leakage **falls to within** 0.07 **of its chance floor by** $\lambda_{\text{adv}} = 0.25$ (0.571 against a floor of 0.5) and goes no lower at any larger weight while AUROC continues to fall from 0.513 to 0.427 as λ_{adv} rises further. The damage continues after invariance is complete.

Second, at the pre-projection backbone, leakage is essentially flat across the whole sweep (0.982 → 0.941) while backbone Mahalanobis AUROC falls 0.749 → 0.475. Domain information remains linearly decodable there, and detection collapses anyway. Because the backbone is a different representation from the 16-dimensional latent, this is the independent replication that the five-detector agreement does not provide.

Third, and most directly, two training objectives traverse the same leakage range with opposite consequences (Section 4.6). Leakage alone therefore does not determine the outcome.

Fourth, in histopathology the two quantities come apart almost completely. In a two-hospital Camelyon17 setting (three seeds, hospital-disjoint slides held out for detection), raising λ_{adv} from 0 to 1 moves all five detectors by 0.12 to 0.22 — Mahalanobis 0.651 → 0.534, cosine 0.785 → 0.589, k-NN 0.794 → 0.648, MSP 0.798 → 0.589, free energy 0.794 → 0.575 — while the hospital probe moves by 0.007 ($0.959 \pm 0.007 \rightarrow 0.952 \pm 0.001$), which is inside its own seed spread at $\lambda_{\text{adv}} = 0$, and in-distribution accuracy is unchanged at 0.995. The adversary reached chance here (final cross-entropy 0.692 against $\ln 2 = 0.693$), so the optimisation pressure was applied; the encoder simply did not become invariant under it. Detection degraded regardless. **This is the clearest dissociation we obtained: the outcome moved roughly twenty-five times as far as the supposed mediator.**

Two qualifications. The detectors fall but remain above chance (0.53–0.65), so this is degradation, not the collapse to chance seen in dermatology, and nothing here bears on the inversion. And these runs come from an experiment that failed its own pre-registered gate — we had required the probe to fall by at least 0.10 before treating the setting as a test of invariance, and it fell by 0.007. That gate failure is precisely what makes the runs informative here: they are a case where invariance was *not* obtained and detection degraded anyway. We report them for this claim only, and the pre-registered verdict on the original question stands as negative (Section 6).

We accordingly report leakage as a measured quantity and never as the explanatory variable, and plot all results against λ_{adv} rather than against leakage.

4.6. The failure is specific to gradient reversal

Adversarial training is one route to invariance among several. Table 2 sweeps four alternatives on the single-encoder backbone: deep CORAL (Sun and Saenko, 2016), maximum mean discrepancy (Long et al., 2015), invariant risk minimisation (Arjovsky et al., 2019) and group distributionally robust optimisation (Sagawa et al., 2020).

Only MMD moves leakage appreciably, and it does so at a price gradient reversal never pays. Escalating the MMD weight from 1 to 10,000 buys 0.13 of leakage (0.967 → 0.836, floor 0.5) at a cost of 0.44 of in-distribution balanced accuracy (0.779 → 0.337), and leaves every detector well above chance (Mahalanobis 0.665, k-NN 0.738, energy 0.694 at the highest weight). IRM reaches the extreme of the same pattern: at weight 100 its balanced accuracy collapses to 0.167 while leakage remains at 0.950. CORAL does not induce invariance at all; at weight 100 it makes the representation *more* domain-separable (leakage 0.998). GroupDRO leaves leakage unmoved at every weight tested.

The contrast is cleanest when task health is held constant rather than leakage, since MMD only reaches comparable leakage by destroying the classifier:

> Gradient reversal reduces leakage 0.970 → 0.763 **at no cost in accuracy** (0.845 → 0.845) and collapses all five detectors to chance. MMD reduces leakage 0.967 → 0.900 at a similarly negligible cost (0.779 → 0.730) and collapses nothing (Mahalanobis 0.813 → 0.716). To go further, MMD must destroy the task. *Out-of-domain set for the MMD figures: [PENDING: R4]; the comparison is only valid if both arms are scored on* `pad_heldout`*.*

The calibration signatures differ in the same direction. As MMD degrades, its out-of-domain ECE *falls* (0.344 → 0.142): the model becomes less confident as it becomes less

accurate, which is the appropriate response. Under gradient reversal out-of-domain ECE triples while accuracy falls. The two objectives move leakage the same way and fail in opposite manners — one degrades honestly, the other is confidently wrong.

We do not claim MMD would be safe at the invariance level gradient reversal reaches; it never got there. What the data establish is that at the leakage levels MMD does reach, it shows none of the failure, while gradient reversal at the same levels has already collapsed.

4.7. It is not specific to the factorized architecture

All results above use a dual-encoder model. Table 3 repeats the sweep with plain single-encoder domain-adversarial training: one backbone, one linear classifier, a gradient reversal layer and a domain head, with no context branch and no orthogonality term.

The cliff reproduces. Every detector collapses at $\lambda_{\text{adv}} = 0.25$ — k-NN 0.944 $\rightarrow$ 0.528, energy 0.911 $\rightarrow$ 0.507, cosine 0.884 $\rightarrow$ 0.509, MSP 0.868 $\rightarrow$ 0.515 — and by $\lambda_{\text{adv}} = 4$ four of the five sit below chance, cosine most clearly at 0.454 ± 0.014. In-distribution accuracy is flat throughout (0.845 $\rightarrow$ 0.832) and cross-domain accuracy falls from 0.390 to a plateau near 0.32. The invisibility, the collapse and the inversion are all properties of domain-adversarial training rather than of the factorized architecture.

A second backbone leaves the question open. Repeating the single-encoder sweep with a ResNet-50 backbone did not settle whether the inversion is specific to EfficientNet-B3, because comparable invariance was never reached. At $\lambda_{\text{adv}} = 2$ the adversary never left chance — best-epoch cross-entropy 0.686 to 0.688 against a threshold of 0.673 — so those three runs are null experiments and we draw no conclusion from them. At $\lambda_{\text{adv}} = 0.5$, where the adversary did learn (best-epoch cross-entropy 0.41 to 0.55), leakage fell only from 0.965 to 0.939 against the 0.742 that EfficientNet-B3 reaches on the same partition, so invariance was not obtained and the backbone question is untested rather than answered in the negative.

On that run the detectors disagree, and we report all of them rather than the one that suits each point. Over the same $\lambda_{\text{adv}} = 0 \rightarrow 0.5$ step, k-NN falls 0.962 $\rightarrow$ 0.563 and cosine 0.922 $\rightarrow$ 0.543, while Mahalanobis moves only to 0.814 and never approaches chance. Three detectors that move together on EfficientNet-B3 separate here. We draw two things from this, and no more. Taking the detectors that do move, the step is the sharpest non-mediation evidence in the paper: the outcome shifts by 0.40 while leakage, the supposed mediator, shifts by 0.026 (Section 4.5). Taking Mahalanobis, the headline detector everywhere else in this paper, nothing happens. **The claim that all five scoring rules collapse together therefore holds on EfficientNet-B3 and does not hold here**, which is an additional reason the backbone question is open.

One difference is informative. The single-encoder model plateaus at leakage 0.76–0.81 and never reaches the 0.571 the dual-encoder attains; more adversarial pressure buys no further invariance. The dual-encoder has somewhere to put the discarded information — a domain probe on its context branch reads > 0.9999 — while the single-encoder must destroy it. The architecture that achieves deeper invariance also suffers the larger collapse (0.427 versus 0.509 at $\lambda_{\text{adv}} = 2$), which is the dose–response relationship seen from a second direction.

4.8. The collapse generalises to a same-modality site shift; the inversion does not

Every result above contrasts dermoscopy with clinical photography, so the construct we call domain is confounded with imaging modality. ISIC 2019 contains a contrast that is not: its `lesion_id` field records the contributing site, with 12,413 images from Hospital Clínic Barcelona (`BCN`) and 10,015 from the HAM10000 collection (`HAM`) — two hospitals, two dermatoscopes, two patient populations, one modality. The two sites are separable from frozen ERM features at balanced accuracy 0.962 ± 0.007, comparable to the 0.970 that ISIC↔PAD reaches, and not for trivial reasons: masking the dermatoscope vignette costs only 0.019, hand-crafted colour statistics reach 0.901 against a 0.876 ceiling on the original pair, and a label-only probe reaches 0.668 (Appendix A.2).

The collapse reproduces. With `BCN` in distribution and a lesion-disjoint 30% `HAM` partition held out, leakage falls $0.963 \pm 0.006 \rightarrow 0.821 \pm 0.030$ at $\lambda_{\text{adv}} = 0.25$ and the detectors with headroom fall through chance: k-NN $0.658 \pm 0.041 \rightarrow 0.486 \pm 0.009$, cosine $0.576 \pm 0.040 \rightarrow 0.477 \pm 0.006$ (3 seeds). Unlike the two multi-domain settings in the limitations, the manipulation took.

The inversion does not. Reversing the roles restores detector headroom — Mahalanobis reads 0.846 ± 0.003 at $\lambda_{\text{adv}} = 0$, essentially the 0.843 of the dermatology sweep — and adversarial training then reduces leakage from 0.95 to 0.77 and drives Mahalanobis to 0.683 ± 0.015 at $\lambda_{\text{adv}} = 1$, where the patient-clustered bootstrap gives [0.666, 0.699] with all three seeds' intervals entirely above 0.5. Detection degrades substantially and stops above chance.

Reading. Baseline detectability is the same (0.846 versus 0.843) and the achieved invariance is the same as the single-encoder dermatology run (0.77 versus 0.763), yet the cross-modality contrast reaches 0.427 while the same-modality one stops at 0.683. Neither a shortage of headroom nor a shortfall in achieved invariance accounts for the gap, so we separate two claims the earlier sections report together. The *collapse* — detection falling to chance as invariance is obtained, invisibly to in-distribution metrics — reproduces between two hospitals. The *inversion* does not: in every setting where we observe it the two domains differ in imaging modality. Whether that reflects a threshold in the size of the shift, something specific to modality gaps, or a property of these datasets, we cannot say.

4.9. The pattern survives a lesion-disjoint split, but the invisibility to accuracy does not

This paper is about shortcut learning, and its main sweep uses an image-level ISIC split in which 3,041 of 5,067 test images (60.0%) share a `lesion_id` with training or validation. We therefore repeated $\lambda_{\text{adv}} \in \{0, 2\}$ with the ISIC split grouped by `lesion_id`, five matched seeds, everything else fixed. Images with a null `lesion_id` (8.23%) each form their own group; no lesion crosses any partition.

Every pre-committed pattern check passes. Leakage still falls ($0.927 \pm 0.018 \rightarrow 0.555 \pm 0.046$), in-distribution ECE is unchanged at 0.181, out-of-domain ECE still rises ($0.219 \rightarrow 0.717$), all five detectors still fall, and the inversion still holds on both out-of-domain sets: Mahalanobis $0.808 \rightarrow 0.403$ on `pad_heldout`, bootstrap CI $[0.380, 0.425]$ with all five seeds individually below 0.5, and $0.717 \rightarrow 0.402$ on Fitzpatrick17k, CI $[0.390, 0.415]$.

In-distribution balanced accuracy does not degrade here either: it reads 0.496 ± 0.016 at $\lambda_{\text{adv}} = 0$ against 0.473 ± 0.023 at $\lambda_{\text{adv}} = 2$, but the paired per-seed difference is -0.023 with a CI of $[-0.063, +0.017]$, which contains zero. We report it as a trend and claim nothing from it.

What the honest split does change is the absolute level, and by far more than λ_{adv} ever does. Removing the lesion overlap costs 0.20 of balanced accuracy at $\lambda_{\text{adv}} = 0$ ($0.692 \to 0.496$), consistent with the companion analysis of lesion-level leakage in this dataset (Tran and TODO, 2027). The leakage this paper is about is therefore worth roughly ten times the largest effect we attribute to the adversarial weight, and the in-distribution accuracies in Table 1 are optimistic by that amount. It does not disturb any comparison across λ_{adv}, all of which are made under one split.

The difference between the two splits is not a change in class balance: the lesion-level test set has the same class distribution to within one image in every class, and reweighting it to the image-level mix leaves every detector value identical to three decimal places.

4.10. The inversion runs through the majority-class centroid

Nevus is 50.8% of the in-distribution set. We pre-registered three criteria for the hypothesis that gradient reversal moves out-of-domain images into the nevus region of the latent space, and all three hold.

(i) Out-of-domain images migrate to the nevus cell. The fraction of `pad_heldout` images whose nearest class centroid is nevus rises from 0.332 at $\lambda_{\text{adv}} = 0$ to 0.601 at $\lambda_{\text{adv}} = 2$ ($\Delta = 0.268$ against $2\,\text{SE} = 0.055$), while the same fraction for in-distribution images is flat ($0.528 \to 0.531$). It tracks the diagnostic shift: across all 27 runs the correlation between this fraction and the number of `pad_heldout` images predicted nevus ($105 \to 424$ of 716) is $\rho = 0.982$.

(ii) The inversion is confined to those images. Split by nearest centroid, AUROC on the nevus-nearest subset falls $0.807 \to 0.282$ while the remainder stays above chance throughout ($0.861 \to 0.644$).

(iii) They move, and the reference does not. Against a class-matched reference — in-distribution images that are themselves nevus — the median squared Mahalanobis distance from out-of-domain images to the nevus centroid falls from 49.8 to 17.5, while the reference distance is unchanged ($7.5 \to 7.4$).

The pattern replicates on the pre-projection backbone, a different representation of 1536 dimensions (nevus-nearest fraction $0.484 \to 0.603$, nevus-nearest AUROC $0.726 \to 0.330$), so it is not a property of the 16-dimensional projection.

What the mechanism does not say. Out-of-domain images never become more typical than real nevi: the distance ratio stays above 1 at every weight ($6.63 \to 2.37$). The inversion is a property of the *pooled* AUROC, which mixes classes. Pooled AUROC equals the class-share-weighted mean of the per-class AUROCs, since AUROC averages over the in-distribution samples — an identity, not a result, which we use only to locate the effect. **The content is the two per-class values:** 0.559 **against nevus and** 0.296 **weighted over the other five.** Out-of-domain images are close to indistinguishable from nevi, whose centroid they are drawn toward, and inverted against everything else; because the majority class is half the data, the pooled score crosses chance. With all eight classes

present the identity holds to machine precision at every weight, both out-of-domain sets and both in-distribution sets (largest residual 2.2×10^{-16}); restricted to the six PAD-shared classes the discrepancy reaches 0.010, the weight of the two omitted classes. It also explains why per-class and pooled AUROC disagree at $\lambda_{\text{adv}} = 0$ (Section 5).

This accounts exactly for a result we had reported and could not previously explain. Reweighting the in-distribution set to PAD's class mix — 12.3% nevus against ISIC's 50.8% — *strengthens* the inversion to 0.259 ± 0.022 (Table 4), because thinning nevus from the reference makes images sitting in the nevus region look comparatively more typical. Reweighting changes only the weights w_c; the model, every image's score and the out-of-domain set are untouched, so no per-class AUROC moves, and evaluating $\sum_c w_c \text{AUROC}_c$ under PAD's mix reproduces the measurement to 8×10^{-7}. A two-group reduction does not, giving 0.306, which is why the magnitude appeared not to follow from the mechanism until the full form was used.

One consequence is worth stating with its scope. **For the pooled score, with the model and out-of-domain set fixed, the inversion is deeper the *less* the in-distribution set is dominated by its majority class**: the two-group reduction $0.559p + 0.296(1 - p)$ for a nevus share p, with the non-nevus classes at ISIC proportions, returns above chance only at $p = 0.776$. That is a statement about one metric under one model, not a rule about deployments.

4.11. What the phenomenon is not

Table 4 in the appendix lists six candidate accounts — memorisation of the adversarial images, class composition, domain specificity, dimensional collapse, latent compression and tail asymmetry — each tested against a prediction fixed before its test, and each refuted, together with a pre-registered synthetic reproduction that failed to reproduce the inversion. We report them because they are the reason the account in Section 4.10 is credible: it is what survived after the alternatives were eliminated, rather than the first story we tried.

5. Discussion

An inversion, not a degradation. The central result is not that out-of-distribution detection weakens under adversarial invariance training but that the ordering reverses: at $\lambda_{\text{adv}} = 2$ a model asked which images are unlike its training data answers with its own test split. Degradation would be unsurprising — a representation stripped of domain information should be less able to recognise a domain — and chance is what that argument predicts, which is what a single-encoder model reaches at $\lambda_{\text{adv}} = 0.25$ (0.525). Going *past* chance is not predicted by it, and is what Section 4.10 accounts for.

The deployment pathway. The results compose into one failure sequence. A model is trained with invariance to acquisition domain, as recommended, and deployed at a new site, where it increasingly predicts the benign majority class — actinic keratosis recall $0.245 \rightarrow 0.016$ — with higher confidence than on its own test data. The monitor that should flag those images does not merely fail to flag them; it ranks them as more routine than familiar ones. Meanwhile the developer's evaluation shows in-distribution accuracy

holding, in-distribution calibration unchanged and the leakage metric improved exactly as intended.

The failure belongs to gradient reversal, not to invariance. Reducing leakage is not sufficient to produce it. MMD reduces leakage along the same axis and collapses nothing at the levels it reaches with the classifier intact, where gradient reversal has already inverted. The two also fail in opposite manners: pushed harder, MMD becomes less accurate *and* less confident and its out-of-domain calibration error falls, while under gradient reversal accuracy falls as confidence rises and calibration error triples. One degrades honestly, the other is confidently wrong. Two bounds follow. We cannot rule out that MMD would behave the same way at the invariance level gradient reversal reaches, because it never got there without destroying the task; and we tested one adversarial formulation, so whether CDAN or adversarial discriminative adaptation behaves like gradient reversal is not established here. The claim is stated for gradient reversal, not for adversarial training in general.

The danger zone is where the method succeeds. Risk scales with *achieved* invariance, not with the nominal weight or the presence of an adversarial term. Where the adversary never learned, leakage stayed at 0.94 across four orders of magnitude of λ_{adv} and the safety failure did not appear either. A weak adversary produces neither debiasing nor harm — so many deployed configurations may be harmless, and a configuration's safety cannot be read off its hyperparameters.

Tolerance without generalisation. Full-spectrum OOD detection (Yang et al., 2023) prescribes that covariate-shifted in-distribution inputs be accepted rather than flagged. Our models achieve that prescription and are unsafe because of it: by $\lambda_{\text{adv}} = 2$ they accept out-of-domain images more readily than their own test data while classifying them at 0.249 balanced accuracy against a floor of 0.167, with higher confidence than in distribution. Acceptance and competence are separate properties, and current protocols evaluate them separately or not at all. Any claim of tolerance to covariate shift should carry cross-domain accuracy on the same inputs, since tolerating inputs a model cannot classify is a failure mode rather than a success.

Two routes to invariance are not interchangeable. Wald et al. (Wald et al., 2021) showed that enforcing multi-domain calibration produces invariance. The converse fails: obtaining invariance adversarially destroys out-of-domain calibration, driving ECE from 0.247 to 0.746. The property one route builds on is the one the other removes. Where invariance is the goal, calibration-based or moment-matching objectives may therefore be preferable, because they reach it through constraints that leave the out-of-domain uncertainty signal intact. Our MMD results are consistent with that, but we have not tested it directly and offer it as a hypothesis rather than a finding.

Recommendation. Report OOD detection and out-of-domain calibration **as a function of achieved invariance** rather than at a single operating point; report cross-domain accuracy alongside any claim of tolerance to covariate shift; and state, when a separate dataset serves as the OOD set, that this measures covariate shift rather than semantic novelty. Practitioners tuning λ_{adv} already train at several values, so this changes what is reported rather than what is run. It converts an invisible failure into a visible number.

Monitor outside the adversarial path. Since the collapse is a property of the representation that received adversarial gradient, one mitigation is to score novelty on a representation that did not. A frozen ImageNet ResNet-50 gives Mahalanobis AUROC 0.988 on Fitzpatrick17k and 0.833 on `pad_heldout`, against the trained model's 0.764 and 0.843 at $\lambda_{\text{adv}} = 0$ and 0.391 and 0.427 at $\lambda_{\text{adv}} = 2$. It never inverts, but **we had pre-registered** 0.90 **in all four cells and** `pad_heldout` **does not reach it**, where the frozen encoder merely matches the untrained model rather than improving on it. A partial mitigation, not a fix; retreating to an earlier layer of the *same* encoder does not work at all (Section 4.5).

The inversion is deepened, not created, in the minority classes. Per-class AUROC, which unlike the pooled value does not depend on the ID class mix, is 0.14–0.39 on `pad_heldout` and 0.10–0.34 on Fitzpatrick17k at $\lambda_{\text{adv}} = 2$ for every non-nevus class. Two are already at or below chance beforehand: on `pad_heldout` at $\lambda_{\text{adv}} = 0$, actinic keratosis is 0.499 and squamous cell carcinoma 0.444. Gradient reversal deepens an inversion that already exists for the rarest and most clinically consequential classes, and the pooled AUROC at $\lambda_{\text{adv}} = 0$ conceals this because nevus is half the in-distribution set.

The mechanism, and what remains open. Gradient reversal removes the acquisition signal by moving out-of-domain images into the region the majority class occupies (Section 4.10) — an efficient way to satisfy the adversary, since the densest part of the latent space is where domain information is hardest to recover, and simultaneously the worst outcome for a distance-based novelty score, which reads density as familiarity. The two objectives are not merely in tension: invariance is obtained by the move that defeats detection. What remains open is why gradient reversal takes this route and moment matching does not, why the synthetic reproduction failed, and whether the majority class is special or merely the densest available — a dataset with two comparably sized dominant classes would separate the last of these.

6. Limitations

Absolute cross-domain performance is poor for every model here. Plain accuracy on `pad_heldout` is 0.245 for the factorized model and 0.391 for the baseline, against a 0.385 majority rate. No model in this study is deployable. Our claim concerns the *effect of* λ_{adv} within a controlled intervention, not the deployability of any particular model, and the deployment pathway above should be read as a mechanism of harm rather than as a description of a fielded system.

The inversion is observed only across imaging modality. ISIC is dermoscopy; PAD-UFES and Fitzpatrick17k are clinical photography, so the main contrast confounds site with modality, and Fitzpatrick17k does not resolve this because it is also clinical photography. The same-modality contrast within ISIC (Section 4.8) separates the two claims: the collapse reproduces between two hospitals, the inversion does not. Every setting in which we observe the inversion is a cross-modality one, and we cannot say whether that reflects the size of the shift, a property of modality gaps specifically, or these particular datasets.

Three attempts to replicate outside dermatology returned no result, for the same underlying reason in different guises: the manipulation did not take, so there was nothing to

measure. On three-hospital Camelyon17 the adversary head never trained; on iWildCam it trained but the encoder never became invariant; and on Camelyon17 restricted to *two* hospitals — the obvious diagnosis, matching the dermatology domain count, with the gradient path verified first — the adversary reached chance ($\ln 2$) while a post-hoc probe still read the hospital at 0.92–0.95, a fall of at most 0.037 against the 0.10 we had pre-committed. The pre-registered stopping rule returns a negative and we stop. **Domain count is not the barrier**, and we cannot say what is; the pattern — an adversary driven to chance while the information it was asked to remove stays linearly decodable — is a known failure mode of adversarial removal (Elazar and Goldberg, 2018). **We report no inversion outside dermatology**; a single-seed sub-chance reading there did not replicate (0.534 ± 0.050 over three seeds). What those runs do support is the dissociation in Section 4.5. Details in Appendix A.1.

Generality beyond dermatology remains untested. **Backbone coverage.** Comparable invariance was not reached with a ResNet-50 backbone, so whether the inversion extends beyond EfficientNet-B3 is untested. Three of the nine runs failed their adversary gate outright and are reported as null experiments rather than negative results. **The objective comparison is bounded.** MMD never reached the invariance level gradient reversal attains; it could only approach it by destroying the classifier. We therefore cannot say what moment-matching would do at matched invariance, only that at the leakage levels it reaches, gradient reversal has already inverted and MMD has not.

Detector non-independence. The five scoring rules are functions of the same 16-dimensional latent. The backbone analysis and the single-encoder replication provide the independent evidence. **Calibration arithmetic.** Out-of-domain ECE is close to mean confidence minus accuracy in this setting; the confidence, accuracy and ECE results are one phenomenon viewed three ways, not three independent findings. **Seeds.** Five seeds at the endpoints, three at interior points. Effect sizes are large relative to seed variance, but formal testing across seven points at $n = 3$ is not claimed.

Splits. The main sweep uses the image-level ISIC split, in which 3,041 of 5,067 test images (60.0%) share a `lesion_id` with training or validation. This inflates in-distribution balanced accuracy by 0.20 ($0.496 \to 0.692$ at $\lambda_{\text{adv}} = 0$), so the absolute values in Table 1 are optimistic by that amount; Section 4.9 reports the lesion-disjoint replication at five seeds, under which every conclusion is unchanged. PAD splits are patient-level, disjoint by patient and by lesion. **Third-domain coverage.** 3,887 of 16,577 Fitzpatrick17k images were obtained; the set is web-scraped under mixed copyright, used for evaluation only, and not redistributed.

7. Conclusion

Under domain-adversarial training, out-of-distribution detection does not merely weaken. It reverses: a model asked which inputs are unfamiliar answers with its own test data and ranks a new clinic's images as the more routine of the two. The reversal appears at the smallest adversarial weight that achieves invariance, holds on a dataset the adversary never saw, reproduces without the factorized architecture, and does not appear under moment-matching objectives that reduce leakage along the same axis.

It is invisible to everything a developer monitors. In-distribution accuracy holds, in-distribution calibration is unchanged and the domain leakage probe improves exactly as intended — under an image-level and a lesion-disjoint ISIC split alike — while confidence on unseen domains overtakes confidence on familiar data and recall for a premalignant class falls to 0.016.

The cause is identifiable: invariance is obtained by moving out-of-domain images into the majority class's region of the latent space, which is exactly the move that makes a density-based novelty score call them familiar. Acting on it does not require the explanation. Reporting out-of-distribution detection, out-of-domain calibration and cross-domain accuracy as functions of achieved invariance costs almost nothing and makes the failure visible.

Data and code availability

https://github.com/TranDucVinh-20225538/CSG-SKin. ISIC 2019 CC BY-NC 4.0; PAD-UFES-20 CC BY 4.0; Fitzpatrick17k annotations CC BY-NC-SA 3.0 with images used for evaluation only and not redistributed.

Declaration of competing interest

The authors declare no competing interests.

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Appendix A. Supporting tables and figures

A.1. External replication attempts

Camelyon17, three hospitals. The adversary head never trained: cross-entropy sat at $\ln 3$ from the first epoch at every λ_{adv} while a post-hoc probe on the same features read 0.94, despite a gradient graph verified correct (head cosine $+1.000$, encoder -1.000).

iWildCam. The adversary did train — best-epoch cross-entropy 3.13 against $\ln 243 = 5.49$ — but the encoder never became invariant: the binary train-versus-OOD site probe moved only from 0.911 to 0.891 across $\lambda_{\text{adv}} \in [0, 10]$ against a floor of 0.5, and no detector collapsed.

Camelyon17, two hospitals. Pre-committed in `PRECOMMIT_CAMELYON2.json` before running. The adversary reached chance (final cross-entropy 0.692 against $\ln 2 = 0.693$, from 0.11 at $\lambda_{\text{adv}} = 0$) and in-distribution accuracy held at 0.995, but the hospital probe fell only $0.959 \pm 0.007 \rightarrow 0.952 \pm 0.001$, inside its own seed spread, so the leakage gate failed and the verdict is negative. The exploratory seeds added afterwards do not reopen it: Mahalanobis on the held-out hospital reads 0.534 ± 0.050 at $\lambda_{\text{adv}} = 1$ over seeds 42–44, above chance, so the single-seed 0.478 did not replicate. These runs appear in Section 4.5 for the non-mediation claim only, where the five detectors move 0.12–0.22 while the probe moves 0.007.

Figure 6: same-modality site shift

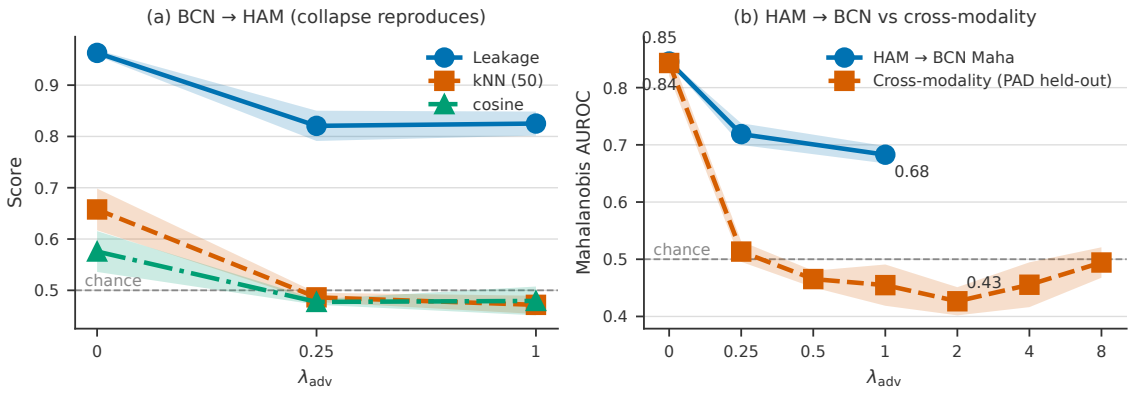


Figure 2: **A same-modality site shift separates the two claims.** (a) Barcelona to HAM10000: leakage falls and the detectors with headroom cross chance. (b) With the roles reversed to restore headroom, Mahalanobis falls from a baseline matching the cross-modality sweep but stops above chance, while the cross-modality sweep continues below it.

A.2. Same-modality site shift: probe validation and the `HAM` direction

The sites are separable, and not for trivial reasons. A logistic probe on frozen ERM features distinguishes `BCN` from `HAM` at balanced accuracy 0.962 ± 0.007 against a floor of 0.5. Masking the dermatoscope vignette reduces this only to 0.943 ± 0.005, so the border is not the signal. Hand-crafted colour and intensity statistics reach 0.901 at best — close to the 0.876 ceiling the same features reach on ISIC↔PAD, so this pair is no more low-level than the original contrast — while sharpness statistics sit at chance (0.532). A probe given only the ground-truth class label reaches 0.668, reflecting the two sites' different class distributions; the feature probe is far above that reference.

Mahalanobis is uninformative in the `BCN`-in-distribution direction. It reads 0.379 at $\lambda_{\text{adv}} = 0$, already below chance before any adversarial training. Reweighting the in-distribution set to the out-of-domain class mix raises it to 0.487, so most of that baseline inversion is class composition — `HAM` is two-thirds nevus and therefore concentrated near a single well-populated centroid — with a residual we attribute to HAM10000's more standardised acquisition and correspondingly lower feature variance. This is why the reversed direction, which restores headroom, carries the scope claim in Section 4.8. In that direction the class-reweighted variant falls from 0.74 to 0.54, and Mahalanobis reads 0.719 ± 0.019 at $\lambda_{\text{adv}} = 0.25$.

A.3. A domain monitor is cheap, and semantic novelty is unaffected

A linear domain head on a frozen ImageNet ResNet-50 that has never seen this data reaches AUROC 0.998; on the trained baseline, 0.997; on the trained EfficientNet-B3 control, 0.991. Detecting the acquisition domain is easy from almost any representation, so retaining a

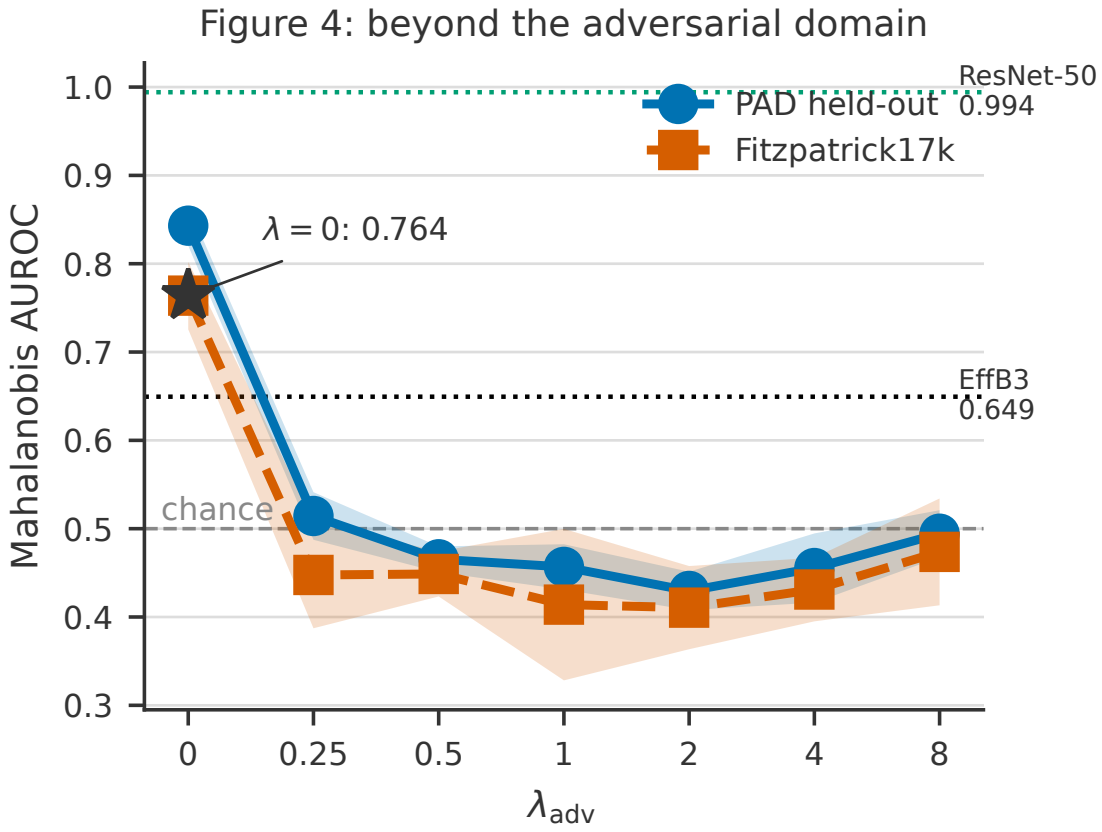


Figure 3: **The collapse is not confined to the adversarial domain.** AUROC against λ_{adv} for the held-out PAD partition and for Fitzpatrick17k, which is absent from every training branch. Horizontal references give the ResNet-50 baseline and the EfficientNet-B3 control on Fitzpatrick; the controlled comparison is the $\lambda_{\text{adv}} = 0$ point of the sweep itself.

monitor is not in itself difficult. What distinguishes the dual-encoder's context branch is *concentration*: a single principal component carries 84.5% of its variance and already separates the domains (AUROC > 0.9999, two discordant pairs in 11.6M), against 0.55–0.70 at $k = 1$ for every other representation tested.

The damage is specific to covariate shift. With the domain held constant and diagnostic classes withheld from training instead, both branches land between 0.6 and 0.8 and the ordering depends on which class is withheld (DF and VASC: lesion 0.648, context 0.641, baseline 0.687; SCC: lesion 0.806, context 0.607, baseline 0.684). No near/far structure organises the difference (Pearson -0.12 against a centroid-similarity index). We report this as a null result.

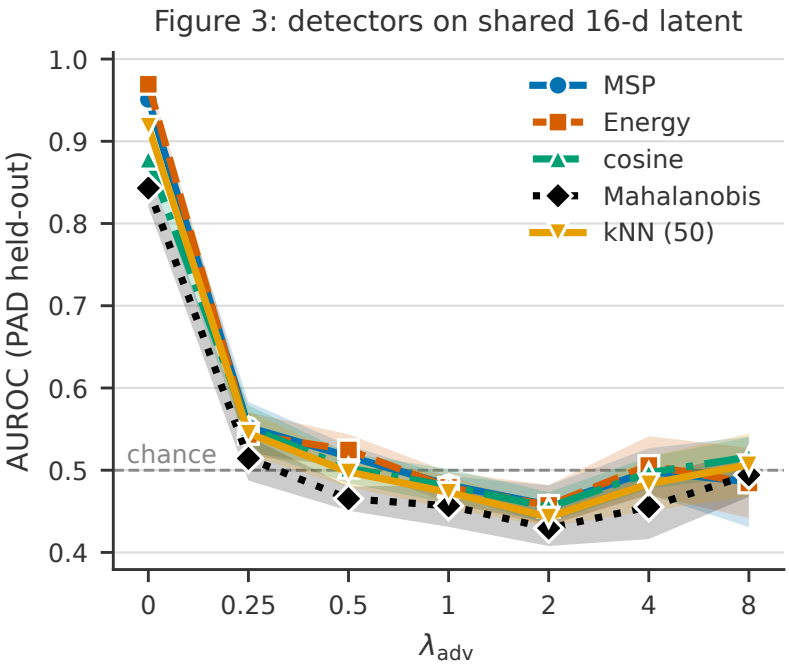


Figure 4: **Every scoring rule moves together.** The five detectors against λ_{adv} on the held-out PAD partition, mean $\pm$ 1 s.d. All five are functions of the same 16-dimensional latent and are therefore not independent evidence; Figure 6 carries the replication on a different representation.

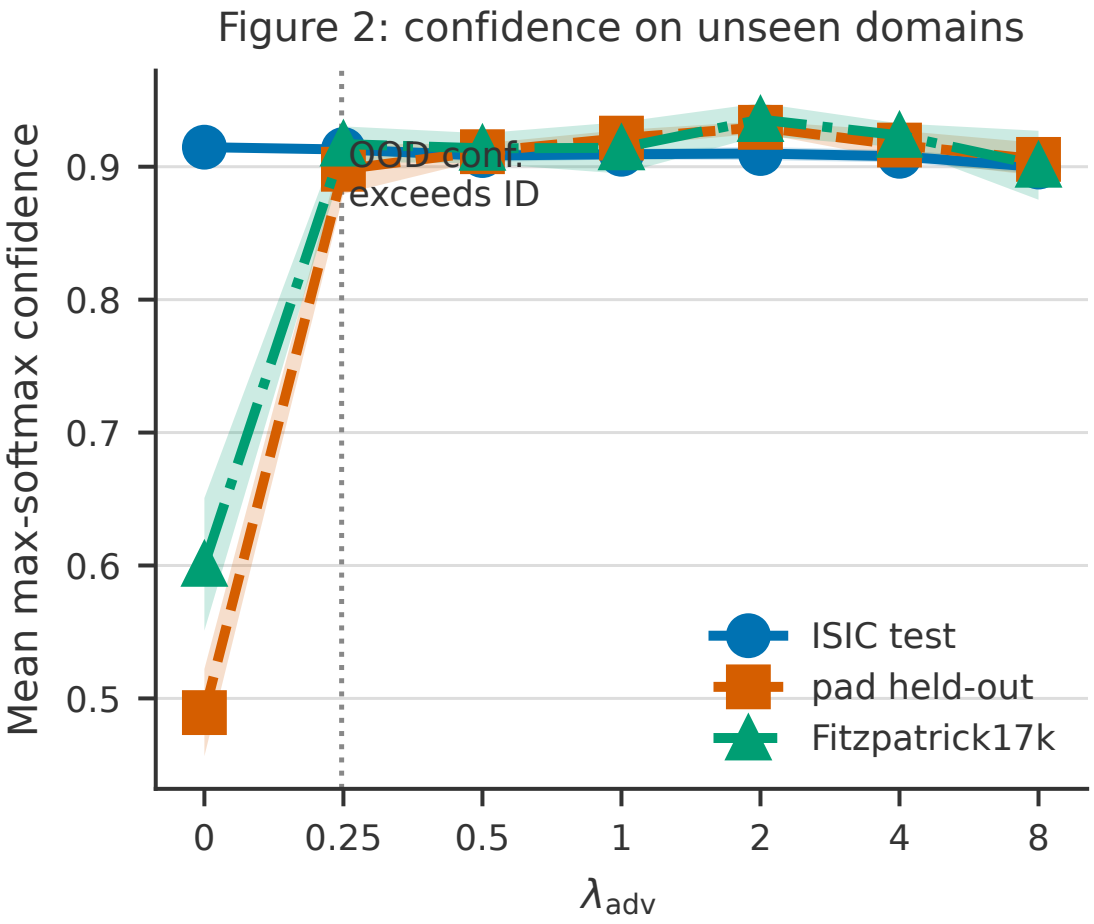


Figure 5: **Confidence on unseen domains overtakes confidence on familiar data.** Mean maximum-softmax probability against λ_{adv} for the ISIC test split, the held-out PAD partition and Fitzpatrick17k. The crossing occurs by $\lambda_{\text{adv}} \approx 0.5$. λ spacing is categorical.

Figure 5: leakage flat, OOD score falls (shared y-range)

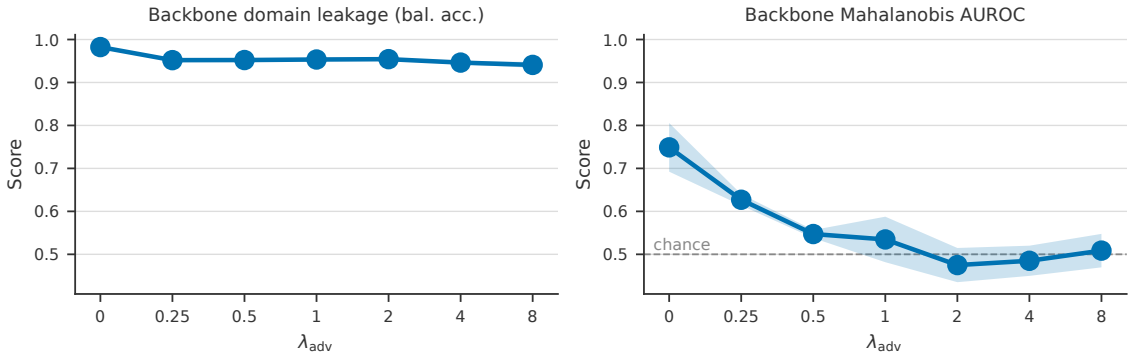


Figure 6: **Leakage does not mediate the collapse.** At the pre-projection backbone the domain probe stays flat across the sweep while Mahalanobis AUROC falls. Domain information remains linearly decodable and detection degrades regardless.

Table 2: Invariance objectives other than gradient reversal, single-encoder backbone, 3 seeds per cell. Leakage is balanced accuracy, floor 0.5. **IRM and GroupDRO use PAD labels** through their environment and group risks; the remaining objectives do not, so their cross-domain numbers are not comparable and the two regimes are kept in separate blocks.

Objective	Weight	Leakage	Maha. AUROC	ID bal. acc	OOD ECE
No access to PAD labels					
ERM[§]	—	0.973	0.838	0.761	—
CORAL	100	0.998	0.999	0.777	—
MMD	1	0.967	0.813	0.779	—
MMD	100	0.900	0.716	0.730	—
MMD	3000	0.893	0.748	0.382	0.142
MMD	10000	0.836	0.665	0.337	0.267
GRL	$\lambda_{adv} = 0.25$	**0.763**	**0.525**	**0.845**[†]	—
Uses PAD labels — not comparable on cross-domain accuracy					
IRM	100	0.950	0.729	0.167	—
GroupDRO	10	0.976	0.868	0.783	—

[†] plain accuracy for the GRL row; see Table 3.

[§] scored on `pad_full` with a uniform optimiser and an ISIC-only loss; the λ_{adv}=0 row of Table 3 uses `pad_heldout`, grouped learning rates and `ignore_index`. They are not the same condition. [PENDING: harmonise or relabel.]

Table 3: Single-encoder DANN on the same data and splits. ID accuracy is plain accuracy; leakage is balanced accuracy, floor 0.5. All five detectors collapse at the smallest non-zero weight, and four of five fall below chance by $\lambda_{\text{adv}} = 4$.

λ_{adv}	n	Leakage	Maha.	k-NN	MSP	cosine	Energy	ID acc	X-dom
0	5	0.970	0.700	**0.944**	0.868	0.884	**0.911**	0.845	0.390
0.25	3	**0.763**	0.525	**0.528**	0.515	0.509	**0.507**	0.845	0.333
0.5	3	0.795	0.510	0.517	0.501	0.493	0.507	0.844	0.328
1	3	0.795	0.548	0.521	0.462	0.482	0.448	0.840	0.334
2	5	0.807	0.509	0.516	0.479	0.468	0.470	0.839	0.325
4	3	0.798	0.502	0.486	0.471	0.454	0.449	0.835	0.320
8	3	0.787	0.520	0.532	0.519	0.490	0.500	0.832	0.321

Table 4: Candidate explanations tested and refuted. Each was a prediction specified before its test.

Explanation	Test	Result
Memorisation of the adversarial images	Hold PAD out of the adversary at patient level	`pad_heldout` 0.427 vs `pad_adv` 0.407; unchanged
Class composition	Restrict ID to the six shared classes; reweight ID to the OOD class mix	0.432 vs 0.427 on `pad_heldout`; reweighting *strengthens* the inversion to 0.259
Domain specificity	Evaluate on Fitzpatrick17k, never in $\mathcal{L}_{\text{adv}}$	0.391 at $\lambda_{\text{adv}} = 2$ against 0.764 at $\lambda_{\text{adv}} = 0$, same model and data
Dimensional collapse	Participation ratio and eigenspectrum across λ_{adv}	4.879 $\rightarrow$ 4.500 while leakage 0.914 $\rightarrow$ 0.553 (same three runs); decoupled
Latent compression	Latent variance and distance to centroid across λ_{adv}	OOD is already 4.5$\times$ tighter at $\lambda_{\text{adv}} = 0$; the adversary *decompresses* it
Tail asymmetry	Recompute AUROC after truncating the pooled upper tail	0.412–0.424 versus 0.427; medians invert
Synthetic reproduction	Pre-registered toy model with heavy-tailed ID structure, grid fixed before fitting	No inversion (λ_{adv}=0: 0.693; λ_{adv}=2: 0.786)